

Unit-1: Analytical Questions

Good

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1. Ciphertext: PHHWPHDIWHUWKH WRJDSOUWD

Step-1: Try small shifts; $P \rightarrow M$ is shift of 3 letters back.

Step-2: Apply shift 3 to every letter.

Key = 3; Plaintext = MEET ME AFTER THE TOGA PARTY

2. SECURITY, Key = 11, $C = (P + 11) \bmod 26$

Step-1: Convert letters to numbers

(S=18, E=4, C=2, U=20, R=17, T=8, Y=24)

Step-2: Add 11 mod 26 to each:

3, 15, 13, 5, 2, 19, 4, 9

Ciphertext: DPNFCTET

3. Step-1: Only 26 possible keys (shifts 0-25).

Step-2: Attacker can test all keys almost instantly.

Time complexity = $O(26) = O(1)$ - formally breakable.

4. Step-1: Key space ($26!$) resists brute force.

Step-2: But each plaintext letter always maps to same ciphertext letter.

Letter frequency patterns of English (E, T, A...) still show through.

enabling frequency analysis regardless of key-space size.

5. GSRH RH Z HVXIVG

Step 1: Recognize mirrored alphabet pattern ($A \leftrightarrow Z, B \leftrightarrow Y \dots$) $\rightarrow$ Atbash

Step 2: Map each letter to its mirror ($G \rightarrow T, S \rightarrow H, R \rightarrow I, H \rightarrow S \dots$).

PT: THIS IS A SECRET

6. Step 1: Assume "x" corresponds to the most frequent English letter "e".

Step 2: Build partial substitution, then test next-frequent letters (T, A, O) against digraphs like "TH", "THE".

Approach: frequency mapping + pattern/digraph matching refined iteratively.

7. Step 1: Same PT letter can map to different ciphertext letters depending on key position.

Step 2: This flattens the ciphertext-frequency distribution.

Result: Single-letter frequency analysis becomes ineffective.

8. CRYPTOGRAPHY, Key $\rightarrow$ KEYKEYKEYKEY

Step 1: Converts letters to numbers, repeat key to match length

Step 2: $C = (P + K) \bmod 26$ for each pair.

Ciphertext = MVWZXMQVYZLW

9. Ciphertext LXFOPVEFRNHR, Keyword LEMON

Step 1: Repeat keyword to length 12: LEMONLEMONLE.

Step 2: $(C - K) \bmod 26$ for each pair

Ciphertext: MVWZXMQVYZLW

Step-1: Short repeating key causes periodic repetition in ciphertext.

Step-2: Kasiski attack finds repeated ciphertext segments, measures distances between them, and takes GCD to estimate key length.

Only key length is known, cipher reduces to multiple Caesar ciphers - Breakable via frequency analysis.

11. Step-1: A truly random key as long as plaintext means no repetition/pattern exists.

Step-2: Every ciphertext is equally likely for any plaintext (perfect secrecy). This becomes a one-time pad - theoretically unbreakable.

12. Step 1: Monoalphabetic: fixed 1:1 mapping $\rightarrow$ low entropy per position, weak confusion.

Step-2: Polyalphabetic: mapping varies with position $\rightarrow$ higher entropy, stronger confusion.

Polyalphabetic ciphers offer greater confusion and resistance to frequency analysis.

13. Step 1: Letters mapped to integers 0-25

Step-2: Encryption/decryption performed as arithmetic mod 26 (e.g., $c = (p+k) \bmod 26$).

Modular arithmetic keeps results within the alphabet range, enabling cyclic shifting.

14. Key $[[1,2],[3,5]]$, plaintext HI ($H=7, I=8$)

Step 1: $C = K \cdot P \text{ mod } 26 \rightarrow [1 \cdot 7 + 2 \cdot 8, 3 \cdot 7 + 5 \cdot 8] = [23, 61]$

Step 2: Reduce mod 26 $\rightarrow [23, 9]$

Ciphertext = XJ

15. Matrix $[[2,4],[2,4]]$

1. $\text{Det} = (2 \times 4) - (4 \times 2) = 0$

2. $\text{gcd}(0, 26) \neq 1$, no inverse exists

Fails - matrix not invertible.

16. 1. Decryption needs inverse of key matrix mod 26

2. Inverse exists only if $\text{gcd}(\text{det}, 26) = 1$
Determinant must be coprime to 26.

17. 1. Caesar shifts one letter at a time.

2. Hill mixes multiple letters via matrix multiply.

3. Matrix mixing gives diffusion $\rightarrow$ Stronger than Caesar.

18. 1. Uniform frequency = no dominant letter bias

2. Matches polyalphabetic/Hill-type mixing, not monoalphabetic
Likely polyalphabetic or Hill cipher.

19. 1. Create random 26-letter permutation as key.

2. Share only via secure offline/encrypted channel.

Never send key in plaintext.

1. Hill Cipher = fixed linear transform per key

2. Same input block $\rightarrow$ same output block

Yes, identical plaintext blocks give identical ciphertext blocks.

21. $C = (7P+3) \bmod 26$

1. Has multiply (7) + add (3) $\rightarrow$ Affine cipher

2. $\gcd(7, 26) = 1$ so invertible

Not simple shift - it's an affine cipher

22. 1. Attacker has plaintext - ciphertext pairs

2. Maps letters to find substitution rule

Enough pairs reveal full key.

23. 1. Vigenere's changes shift per position (confusion)

2. But doesn't spread letters across positions (no diffusion)

Good confusion, weak diffusion.

24. 1. Search ciphertext for frequent 3-letter patterns

2. Map matches to T, H, E

Speeds up key recovery

25. 1. Rank by key space + frequency resistance

2. Caesar (26 keys) $\angle$ Mono (26! but freq-weak) $\angle$ Vigenere (freq-resistant, Kasiski-weak) $\angle$ Hill (matrix diffusion)

weakest $\rightarrow$ Strongest: Caesar $\angle$ Monoalphabetic $\angle$ Vigenere $\angle$ Hill

26. RETREATIMMEDIATELY, 3-row
1. Build zigzag rows, read row-wise
 2. Correct output has 18 letters; given has only 17
 3. Given cipher incorrect - a letter missing / misplaced

27. ATTACKATDAWN, 4 columns
1. $12 \text{ letters} \div 4 = 3 \text{ rows}$ exactly, no padding
 2. Match known plaintext to ciphertext blocks to find column order
Table filled row-wise, no padding needed

- 28.
1. Repeated key letters cause same rank
 2. Ties broken by left-to-right order
Ambiguity resolved by first-occurrence order.

29. SECRETMISSION, 2-row
1. Row 0 (even) = S, E, M, S, I, N; Row 1 (odd) = E, R, T, I, S, O
 2. Correct = SCEMSINERTISO
Given ciphertext differs - Zigzag applied incorrectly

- 30.
1. 24 divisible by both 4 & 6 evenly
 2. No padding needed
 3. Dimensions: 6×4 or 4×6 .

31. ZEBRA (5 letters), ciphertext (15 letters)
1. $15/5 = 3$ rows exact
 2. Rank ZEBRA alphabetically $\rightarrow$ Column order
Reordering columns via key rank recovers plaintext.

1. Transposition rearranges positions only
 2. Letter / digram frequencies unchanged
- Frequency analysis fails - stats stay same.

- 33.
1. Happens with repeated / symmetric plaintext
 2. Different key permutations can give same column order
- Occurs due to plaintext redundancy.

34 Length 21

1. Try candidate rail counts (2,3,4...)
 2. Reconstruct and check readability
- Systematic trial-and-check by rail count.

- 35
1. Each 5-length key $\rightarrow 5! = 120$ permutations
 2. Combined $= 120 \times 120 = 14,400$
- Higher permutations $\rightarrow$ stronger brute-force resistance.

- 36.
1. Fixed padding char creates predictable pattern
 2. Attacker uses it to locate block boundaries
- Predictable padding weakens security

37 Length 17 (prime)

1. Only divisors are 1 and 17
2. Forces padding to reach composite length
3. Prime length restricts valid grid dimensions.

- 38.
1. Wrong fill - direction assumption misaligns letters
 2. Output becomes garbled produces meaningless decryption.

- 39.
1. In 3-rail, first/last letters always on top rail
 2. No crossover affects boundary positions Their position stays fixed mathematically.

- 40.
- Length 23, key 6
1. $23 \div 6 = 3$ remainder 5 $\rightarrow$ 4 rows needed = 24 cells
 2. Empty cells = $24 - 23 = 1$
1 empty cell causes ambiguity in decryption.

- 41.
- PLANT & EARTH (5x5)
1. Each has $5! = 120$ permutations
 2. Combined = 120×120
Total = 14,400 permutations.

- 42.
1. Try single-transposition decrypt first
 2. If unreadable, apply second reordering If pattern persists, likely double transposition.

- 43.
1. Repeated letters tie in ranking
 2. Ties broken by original position Column order becomes position - dependent.

1. Fixed-width repetition suggests grid structure.
2. Columnar transposition naturally produces this points to Columnar transposition (5 columns).

- 45.
1. Equal row lengths imply divisor relationship
 2. Check length/rails for even split. Rails = divisor of message length giving equal rows.

- 46.
1. Divisible key lengths cause overlapping structure.
 2. Creates recoverable periodicity weakens double transposition's security.

- 47.
1. Padding added when length isn't multiple of columns.
 2. Extra chars increase total length.
Occurs when $\text{plaintext} \bmod \text{key-length} \neq 0$.

- 48.
- 6 Columns
1. Permutations = $6! = 720$
 2. Small enough for modern brute-force 720 keys - easily broken

49. CRYPTOGRAPHY VS CRYPTANALYSIS

1. 2-rail: row 0 = C, Y, T, G, A, H;
row 1 = R, P, O, R, P, Y

2. Combined = CYTGAIHRPORPY matches given

Confirmed 2-rail fence encryption.

50.

1. Numeric keys give direct column order, no ranking needed

2. Removes the ambiguity from repeated letters

Advantage: clearer ordering;

Disadvantage: still brute-forceable, less memorable.